Causal Effects of Mining Costs on Bitcoin Prices

Evidence from Energy-Based I.V.s

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Abstract

This paper explores the causal effect of Bitcoin mining costs on Bitcoin prices. To address endogeneity between price and mining activity, an instrumental variables approach is used, where mining costs are instrumented with lagged energy prices and estimated using two-stage least squares. The analysis is conducted using weekly data on Bitcoin prices, mining costs, and macroeconomic controls over the sample period.

Introduction

Bitcoin is often viewed as purely a speculative asset driven by demand, but this ignores its cost of production. Bitcoin is produced through mining, a process which involves specialized computers competing to solve cryptographic problems; the first to solve the problem adds a new "block" to the blockchain and receives a reward in newly created Bitcoin. As more miners participate, the network automatically adjusts the difficulty of these problems, making mining more costly. Assuming the minimum required participation is met, Bitcoin has a defined and trackable rate of additional supply. Roughly every four years, the reward that miners receive for adding a new block is cut in half, reducing the number of new Bitcoin entering the market and increasing the effective cost of producing each additional Bitcoin. The mining process requires significant computational power, which in turn takes electricity and hardware investment. This has led to the idea that Bitcoin may behave like a commodity, where production costs play a role in determining price.

However, estimating a relationship between the price of Bitcoin and the cost of mining Bitcoin is complicated by endogeneity, as mining costs are not independent of Bitcoin prices. When prices rise, mining becomes more profitable, which increases mining participation, raises network difficulty, and therefore increases costs. To address this issue, this paper uses an instrumental variables approach, using energy price shocks as an instrument for mining costs. Energy prices directly affect the costs of production, and the identification strategy relies on an assumption that energy prices are exogenous to Bitcoin-specific demand shocks, $Cov(Zt, ut) = 0$. This model is estimated using two-stage least squares. The goal of this paper

is to determine if mining costs have a causal effect on Bitcoin prices, and, more broadly, if Bitcoin behaves like a production-based asset or a purely speculative investment.

Data

The dataset consists of 519 weekly observations of Bitcoin prices, mining cost proxies, energy prices, and macroeconomic controls from 2014 through 2024. All variables are constructed from daily data and aggregated to weekly frequency by taking the average of daily observations within each week. The weekly time index is defined using calendar weeks, ensuring consistent alignment across the series. The final dataset contains observations after removing missing values.

Bitcoin price data is obtained from Yahoo Finance (BTC-USD). The variable *btc_price* is defined as the average of daily closing prices within each week. The dependent variable, *ln_btc*, is the natural logarithm of this weekly average price, allowing regression coefficients to be interpreted as elasticities.

$$B_1 = \frac{\partial ln(BTCt)}{\partial ln(MiningCost(t-1))}$$

The key explanatory variable is a proxy for the cost of mining Bitcoin. Mining costs are not directly observable, so they are approximated using blockchain difficulty and energy price inputs. Weekly mining difficulty is constructed by averaging daily observations of network difficulty. In addition, the Bitcoin block reward is incorporated based on the protocol's halving schedule, where the number of Bitcoin rewarded is halved around every 4 years. The mining cost proxy is proportional to network difficulty and energy input costs, capturing the marginal cost of producing Bitcoin. The variable *ln_mining_cost* is defined as the natural logarithm of this proxy, and *ln_mining_cost_lag1* represents its one-period lag to address simultaneity between mining cost and Bitcoin price.

To identify causal effects, exogenous variation in energy prices is used as an instrument for mining costs. Crude oil prices (WTI, ticker CL=F) are obtained from Yahoo Finance and aggregated to weekly averages. The variables *ln_oil_lag1* and *ln_oil_lag2* are defined as the first and second lags of the logarithm of weekly oil prices. Oil prices are assumed to be a relevant instrument because they directly affect the cost of electricity and mining, while being plausibly exogenous to Bitcoin-specific demand shocks.

Several macroeconomic controls are included to account for broader financial conditions. The VIX index, a measure of market volatility, and Treasury yields (proxying interest rates), are obtained from financial data sources and aggregated to weekly averages, then lagged by one period (*vix_lag1*, *treasury_lag1*). S&P 500 returns are constructed from daily index values as

percentage changes, averaged to the weekly level, and lagged (*sp500_return_lag1*) to capture general market performance and risk sentiment.

All variables are transformed using weekly aggregation and lagged structures to ensure temporal ordering. The log transformation allows coefficients to be interpreted as elasticities and weekly aggregation reduces high-frequency noise.

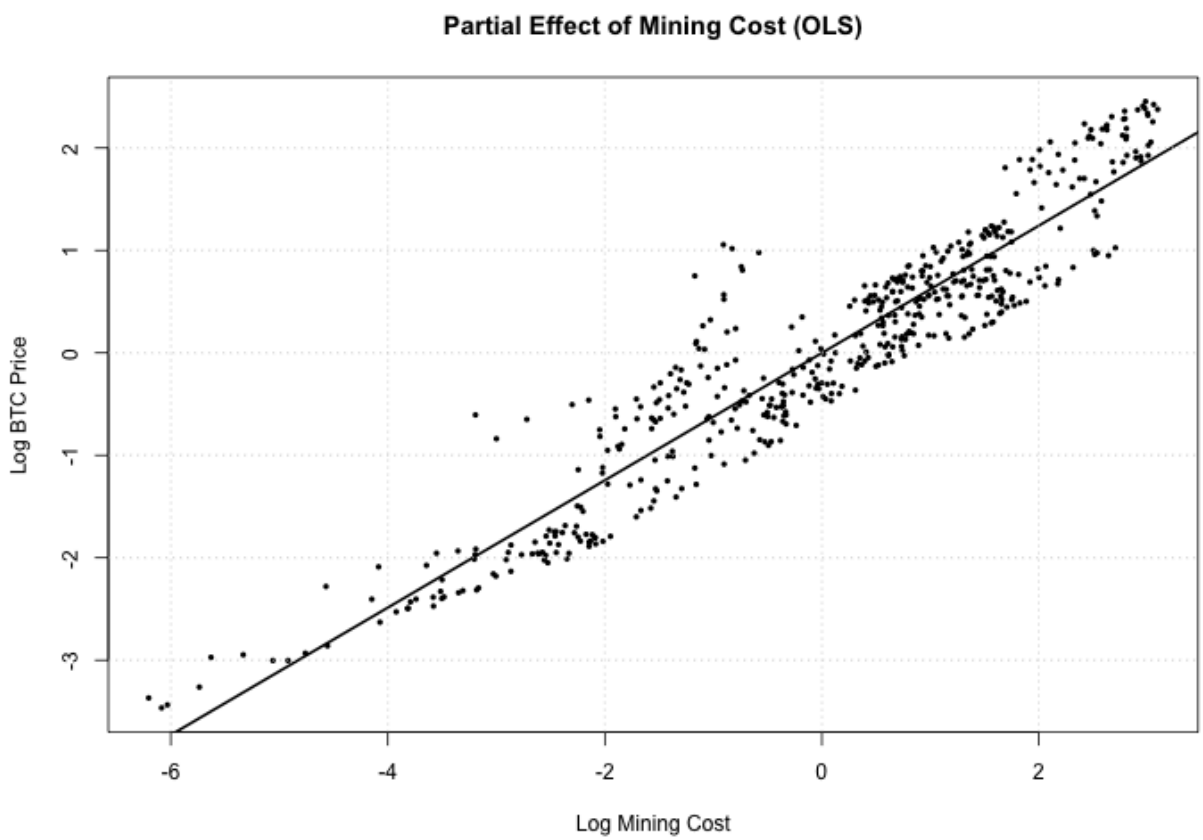
Methods

The empirical model is

$$ln(BTCt) = \beta 0 + \beta 1 ln(MiningCost(t-1)) + \gamma 1 sp500(t-1) + \gamma 2 vix(t-1) + \gamma 3 treasury(t-1) + ut$$

Standard errors are computed using Newey–West corrections to account for both heteroskedasticity and serial correlation in the weekly time series. This is important given the persistence typically observed in financial variables. The bandwidth parameter is set to 6 lags, given the weekly frequency of the data and the persistence that is commonly observed in financial variables.

$$L = [4(\frac{T}{100})^{\frac{2}{9}}]$$



While the OLS results suggest a strong economic relationship, they should be interpreted with caution. Mining costs are likely endogenous, as increases in Bitcoin prices can themselves drive increases in mining activity and network difficulty.

Ordinary Least Squares Regression is likely to be biased due to our exogeneity condition:

$$Cov(ln(MiningCost(t-1)), ut) \neq 0$$

First Stage (Instrument Relevance):

$$ln(MiningCost(t-1)) = \pi 0 + \pi 1 ln(Oil(t-1)) + \pi 2 ln(Oil(t-2)) + \delta 1 sp500(t-1) + \delta 2 vix(t-1) + \delta 3 treasury(t-1) + v(t)$$

First Stage Regression

	Dependent variable:
	Log Mining Cost (t-1)
Log Oil (t-1)	2.892***
	(0.827)
Log Oil (t-2)	0.944
	(0.908)
S&P 500 Return (t-1)	27.032***
	(5.150)
VIX (t-1)	0.171***
	(0.028)
Treasury Yield (t-1)	9.630***
	(1.853)
Constant	10.545***
	(3.527)
Observations	519
R^2	0.615
Adjusted R^2	0.611
Residual Std. Error	1.640 (df = 513)
F Statistic	163.653*** (df = 5; 513)
Note:	*p<0.1; **p<0.05; ***p<0.01

The first-stage regression provides strong evidence that the instrumental variables satisfy the relevance condition. The coefficient on $ln(Oil(t-1))$ is positive and statistically significant $\widehat{\pi 1} = 2.892,\ p < 0.001$, indicating that increases in energy prices are associated with higher mining costs. While the second lag of oil prices is not individually significant, it contributes to the joint explanatory power of the instruments. This is confirmed by the first-stage F-test of joint significance, F = 87.63 ($p < 0.001$), allowing us to reject the null hypothesis that instruments are weak, as F > 10.

Relevance:

$$Cov(ln(Oil), ln(MiningCost)) \neq 0$$

Exogeneity:

$$Cov(ln(Oil), ut) = 0$$

Proceeding to two stage least squares regression:

$$ln(BTCt) = \beta 0 + \beta 1\, ln(\widehat{MiningCost}(t-1)) + \gamma 1 sp500(t-1) + \gamma 2 vix(t-1) + \gamma 3 treasury(t-1) + \varepsilon t$$

Regression Results

	Dependent variable:	
	Log Bitcoin Price	
	OLS	*instrumental variable*
	OLS	instrumental variable
	(1)	(2)
Log Mining Cost (t-1)	0.622***	0.605***
	(0.023)	(0.059)
S&P 500 Return (t-1)	-1.030	-0.668
	(1.154)	(1.712)
VIX (t-1)	-0.020***	-0.018
	(0.008)	(0.015)
Treasury Yield (t-1)	-2.019***	-1.759
	(0.564)	(1.443)
Constant	-9.527***	-9.101***
	(0.606)	(1.328)
Observations	519	519
R^2	0.926	0.925
Adjusted R^2	0.925	0.925
Residual Std. Error (df = 514)	0.420	0.421
F Statistic	1,601.106*** (df = 4; 514)	
Note:		*p<0.1; **p<0.05; ***p<0.01

The baseline OLS estimates indicate a strong positive relationship between lagged mining costs and Bitcoin prices. The estimated coefficient on $ln(MiningCost(t-1))$ is $\widehat{\beta}_1 = 0.622$ (Newey–West SE = 0.0228), with a t-statistic of 27.22 and a p-value below 0.001, providing overwhelming evidence against the null hypothesis $H0: \beta 1 = 0$.

The IV estimates confirm a positive and statistically significant causal effect of mining costs on Bitcoin prices. The coefficient on $ln(MiningCost(t-1))$ is $\widehat{\beta}_1^{IV} = 0.605$ (Newey–West SE = 0.0459), with a t-statistic of 13.16 and a p-value below 0.001.

The IV estimate is slightly smaller than the OLS estimate, which is consistent with the possibility of upward bias in OLS due to endogeneity between mining costs and Bitcoin prices. Because the IV strategy isolates variation in mining costs driven by exogenous oil price shocks, this estimate can be interpreted as causal under the assumption that $Cov(Z, ut) = 0$. The similarity between the IV and OLS estimates (0.605 vs. 0.622) suggests that any endogeneity bias in OLS is modest in magnitude. This conclusion is supported by the Wu–Hausman test (p=0.323), which fails to

reject the null hypothesis of exogeneity, and implies that while IV provides a causal interpretation, the OLS estimates are not materially biased. The Sargan test of overidentifying restrictions ($p = 0.124$) also fails to reject the null hypothesis, providing additional support for validity of instruments.

The control variables behave largely as expected. Higher market volatility is associated with lower Bitcoin prices, consistent with Bitcoins volatility and price instability . Treasury yields enter with a negative coefficient, suggesting that higher interest rates reduce the relative attractiveness of Bitcoin for investors. S&P 500 returns are not statistically significant, indicating that short-run market movements do not strongly predict Bitcoin prices in this specification.

Conclusion

The empirical results support a model in which Bitcoin prices are partially responsive to exogenous variation in production costs. Because the identifying variation comes from energy price shocks, this estimate can be interpreted as a local average treatment effect, reflecting the response of Bitcoin prices to changes in mining costs induced by exogenous energy fluctuations. Using our $\widehat{\beta}_1^{IV}$ estimate 0.605, expressed as elasticity, a 1% increase in mining costs is associated with approximately a 0.60% increase in Bitcoin prices**,** holding other factors constant.

This distinguishes Bitcoin from assets whose prices are determined solely by expectations or speculative demand, as the results provide evidence that Bitcoin exhibits characteristics of a production-based asset, with mining costs acting as a meaningful determinant of price. However, the magnitude of elasticity is less than 1, implying that mining costs do not uniquely determine Bitcoin prices. While energy markets can influence Bitcoin prices through mining costs, they do not fully anchor price behavior.

For traders and investors, the results imply that exogenous energy shocks may shift the distribution of future Bitcoin prices through their impact on mining costs. However, because $\widehat{\beta}_1 < 1$ and demand-side variation remains dominant, energy prices should be viewed as an informative but incomplete predictor of price movements.